Ericsson PE Node WHI-PE-14 Site Whitefield East POP Hardware/Environmental Fault ALARM_4265 SOP

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_4265

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and remediate a hardware/environmental fault on the Ericsson PE node WHI-PE-14 located at the Whitefield East POP. The failure is characterized by ASIC memory parity errors and line card crashes, resulting in the loss of power supply redundancy. This condition triggers the critical alarm ALARM_4265. The underlying protocol and hardware mechanics that contribute to this failure domain are rooted in the PE node's architecture, which relies on a distributed control plane to manage traffic flows. In the event of a hardware failure, the control plane's ability to maintain consistency and redundancy is compromised, leading to the observed symptoms. Furthermore, the Ericsson PE node's reliance on ASIC memory for traffic processing and forwarding exacerbates the impact of memory parity errors, which can cause line card crashes and subsequent loss of power supply redundancy.

Pre-requisite Checks

- pre-requisite: Verify redundant link status on the PE node WHI-PE-14
- pre-requisite: Check the power supply redundancy status on the PE node WHI-PE-14
- pre-requisite: Ensure remote hands are dispatched to the Whitefield East POP for physical access

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash due to hardware failure	High
Loss of power supply redundancy	Medium

Diagnostic Commands & Expected Outputs

```
Command: show hardware alarm all, Expected Output: ALARM_4265: ASIC memory parity error detected on PE node WHI-PE-14, Description: This command displays all hardware alarms on the PE node, including the ALARM_4265 triggered by the ASIC memory parity error.

Command: show linecard status, Expected Output: Line card crashed on PE node WHI-PE-14, Description: This command displays the status of the line cards on the PE node, including any crashes or errors.
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Command: show redundancy status, Expected Output: Power supply redundancy lost on PE node WHI-PE-14, Description: This command displays the redundancy status of the power supplies on the PE node, including any losses or failures.
```

Resolution Steps

Step 1: Step: Replace the failed line card, Command: replace linecard , Description: This command initiates the replacement of the failed line card on the PE node. The slot number of the failed line card must be specified.

Step 2: Step: Reconfigure power supply redundancy, Command: configure redundancy , Description: This command reconfigures the power supply redundancy on the PE node. The redundancy type must be specified according to the PE node's configuration.

Step 3: Step: Verify ASIC memory parity error status, Command: show hardware alarm all, Description: This command verifies the status of the ASIC memory parity error on the PE node. If the error persists, further troubleshooting may be required.

Post-Check Verification

```
Command: show hardware alarm all, Expected Output: No ALARM_4265: ASIC memory parity error detected on PE node WHI-PE-14, Description: This command verifies that the ALARM_4265 has been cleared and the ASIC memory parity error is no longer detected on the PE node.

Command: show linecard status, Expected Output: Line card status normal on PE node WHI-PE-14, Description: This command verifies that the line card status is normal on the PE node.

Command: show redundancy status, Expected Output: Power supply redundancy normal on PE node WHI-PE-14, Description: This command verifies that the power supply redundancy is normal on the PE node.
```

Roll-back Procedures

- Step: Restore the original line card configuration, Command: undo replace linecard , Description: This command restores the original line card configuration on the PE node. The slot number of the replaced line card must be specified.

- Step: Revert power supply redundancy configuration, Command: undo configure redundancy , Description: This command reverts the power supply redundancy configuration on the PE node to its original state.